

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 04



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Header

Topic

TCP Connections



Topics to be Covered



Topic

TCP Connections

Topic

TCP Data transfer

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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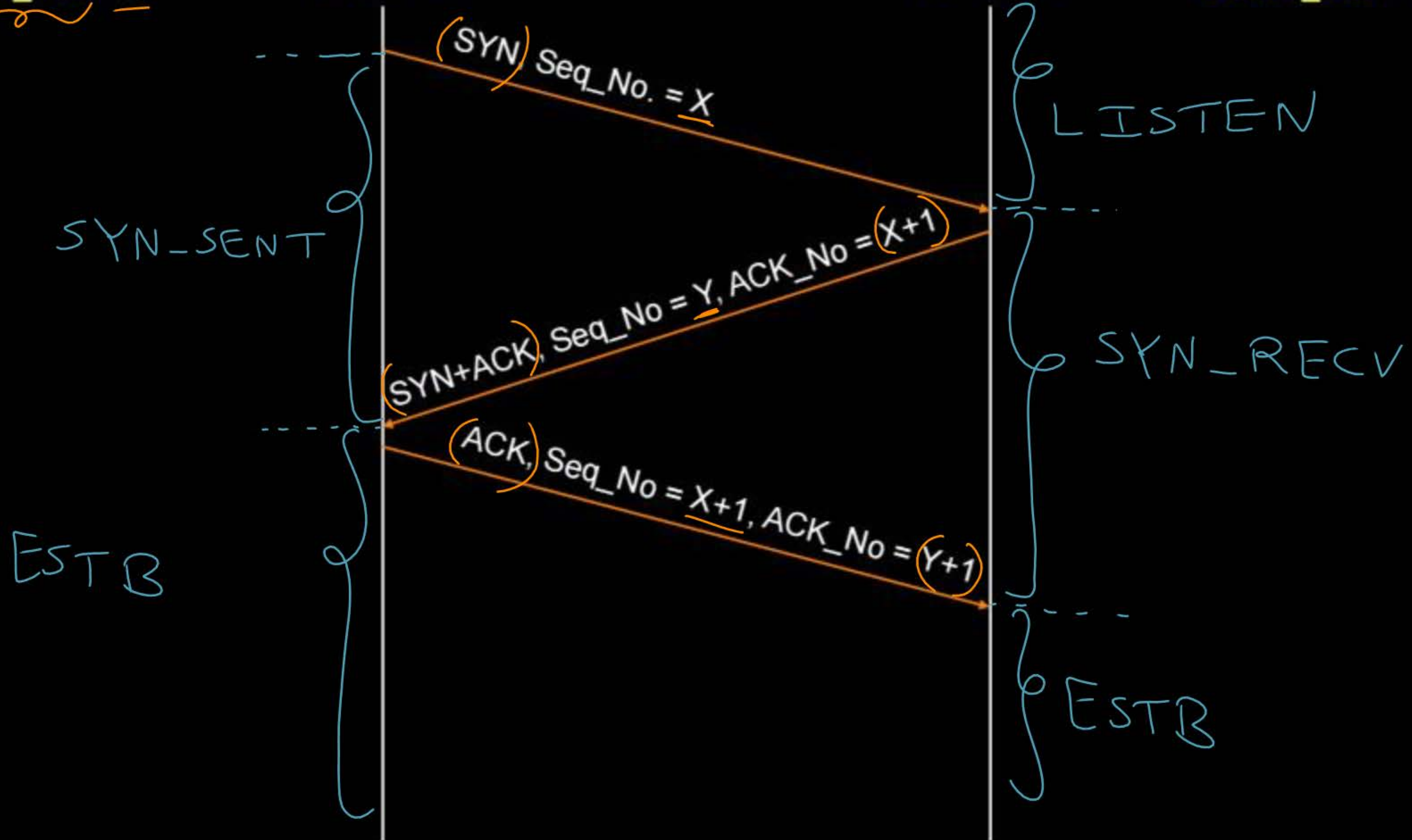


Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y



[MSQ]

[GATE-2025][1 Mark]



#Q. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

☒ A

P1: SYN = 1

☒ B

P2: SYN = 1, ACK = 1

☐ C

P2: SYN = 0, ACK = 1

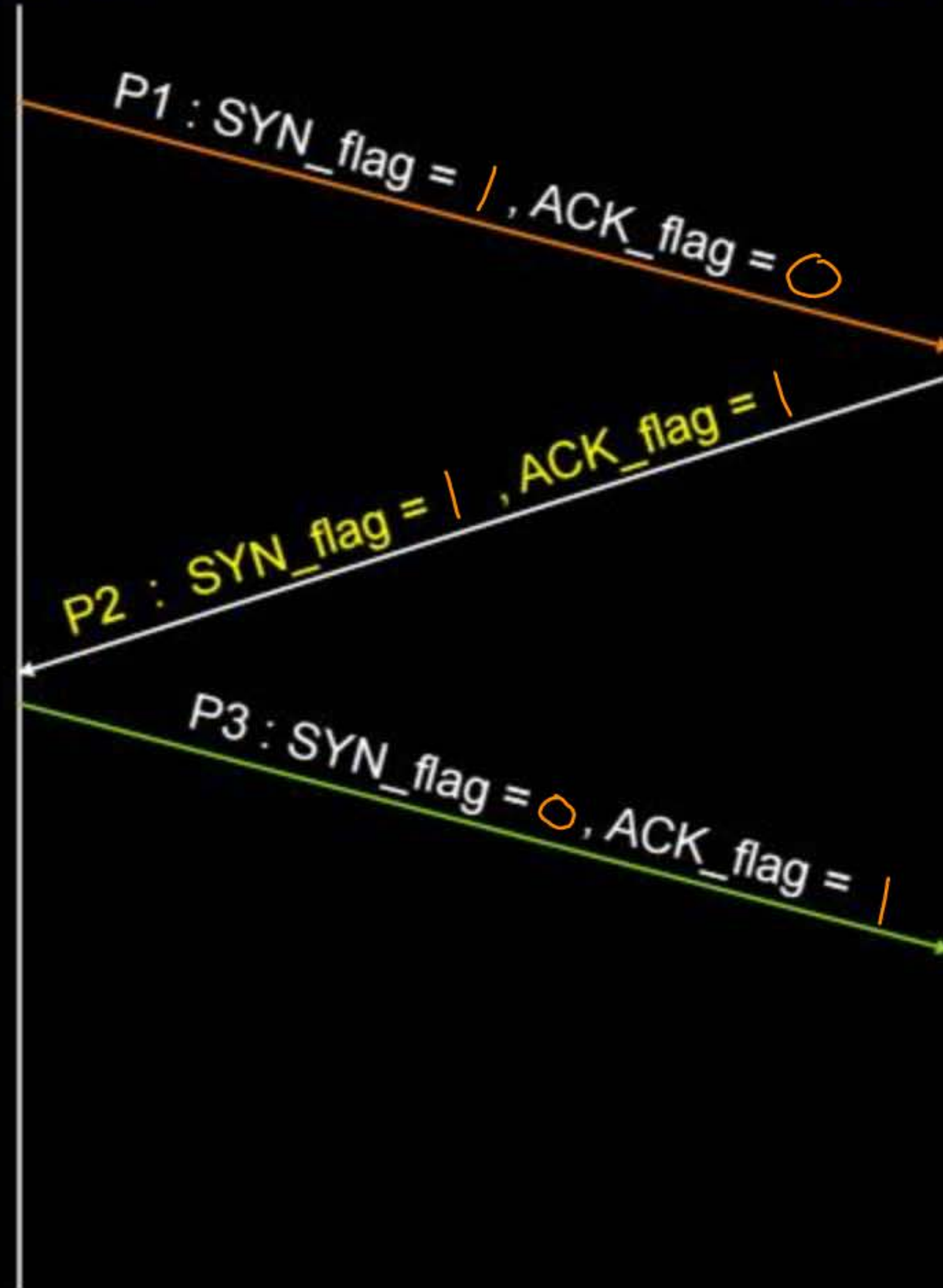
☐ D

P3: SYN = 1, ACK = 1

Ans: A & B

TCP Client

TCP Server



[MSQ]

[GATE-2024][1 Mark]



#Q. TCP client P successfully establishes a connection to TCP server Q . Let N_P denote the sequence number in the SYN sent from P to Q . Let N_Q denote the acknowledgement number in the SYN ACK from Q to P . Which of the following statements is/are CORRECT?

- ☒ A The sequence number N_P is always 0 for a new connection FALSE
- ☒ B The sequence number N_P is chosen randomly by P TRUE
- ☒ C The acknowledgement number N_Q is equal to N_P FALSE
- ☒ D The acknowledgement number N_Q is equal to $N_P + 1$ TRUE

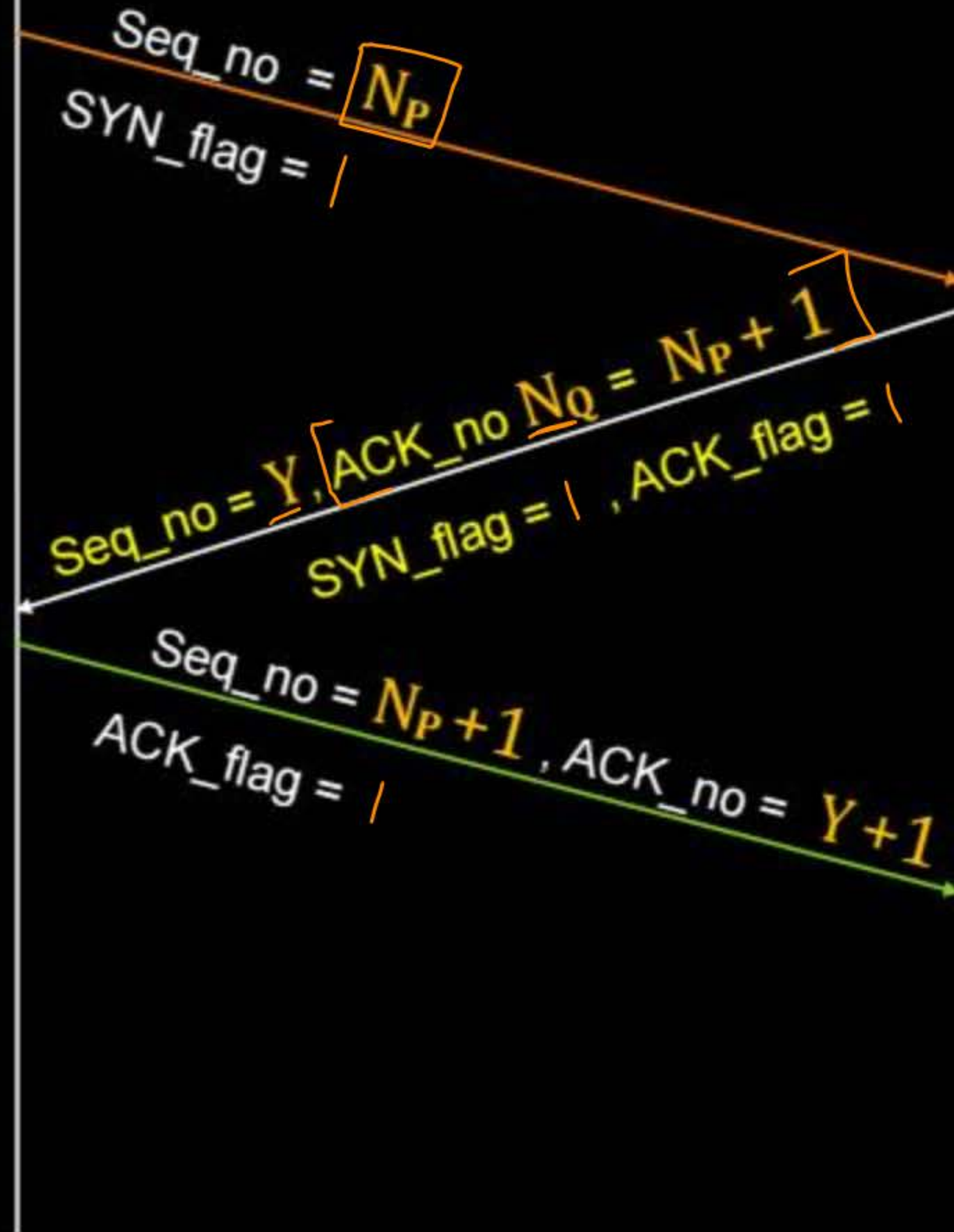
Ans: B & D

Client_ISN = N_P

TCP Client P

TCP Server Q

Server_ISN = Y



CASE I:

Client_ISN = X

Server_ISN = Y

TCP Client

TCP Server

SYN_SENT

TO

LISTEN

SYN_RECV

ESTB

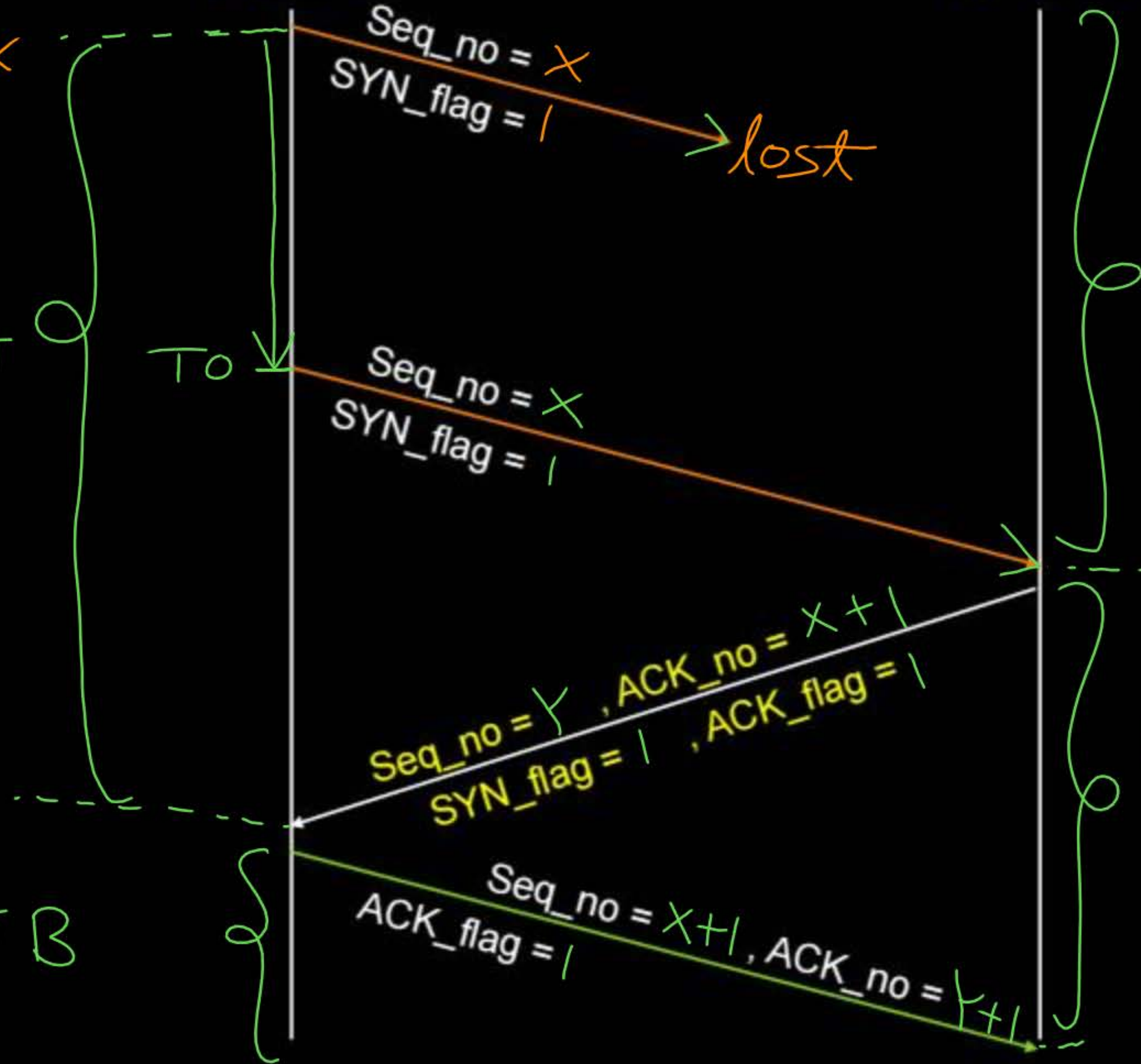
Seq_no = X
SYN_flag = 1

lost

Seq_no = X
SYN_flag = 1

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

Seq_no = X+1, ACK_no = Y+1
ACK_flag = 1



CASE II :

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y

Seq_no = X
SYN_flag = 1

LISTEN

lost
Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

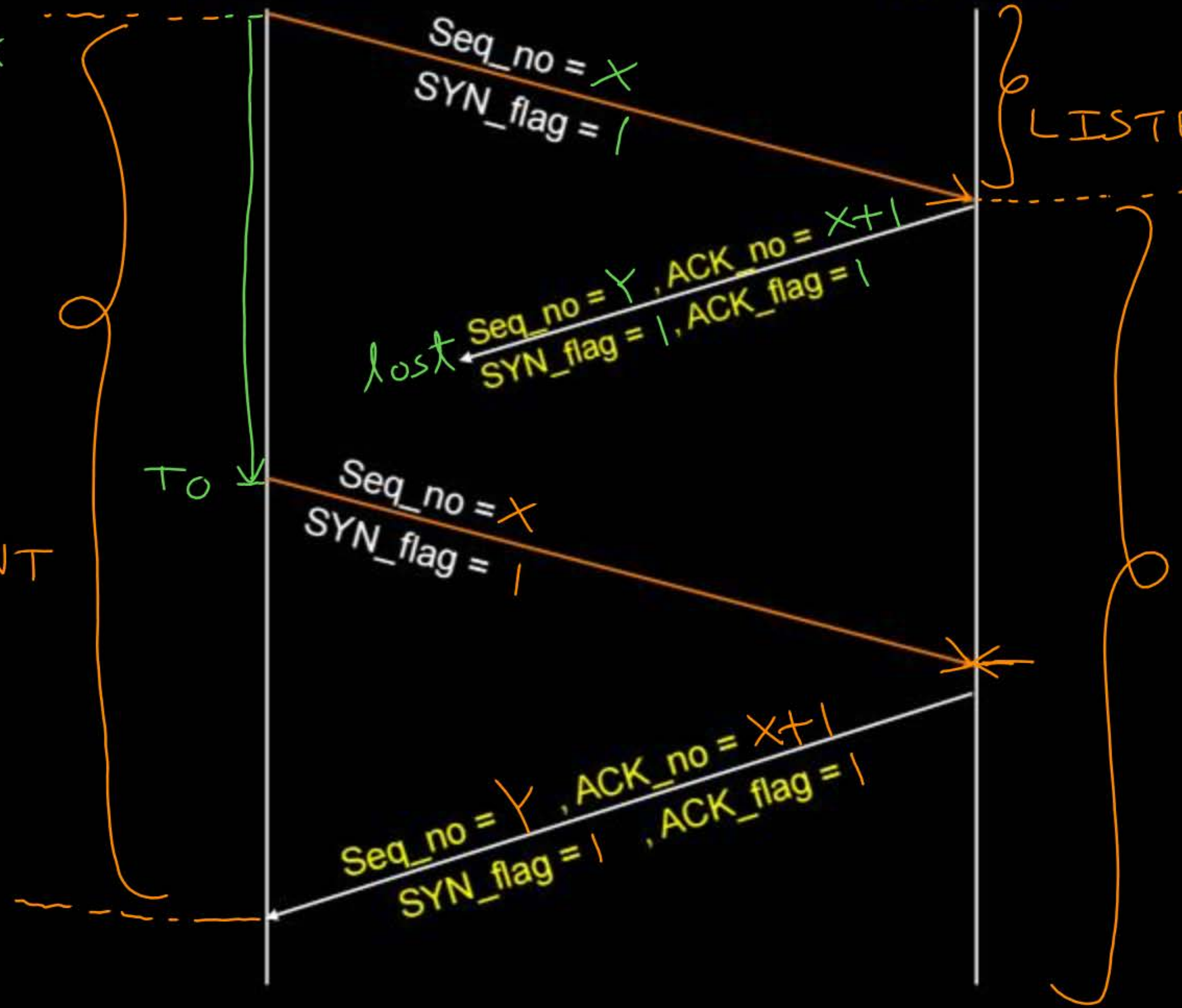
To

Seq_no = X
SYN_flag = 1

SYN-SENT

SYN-RECV

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1



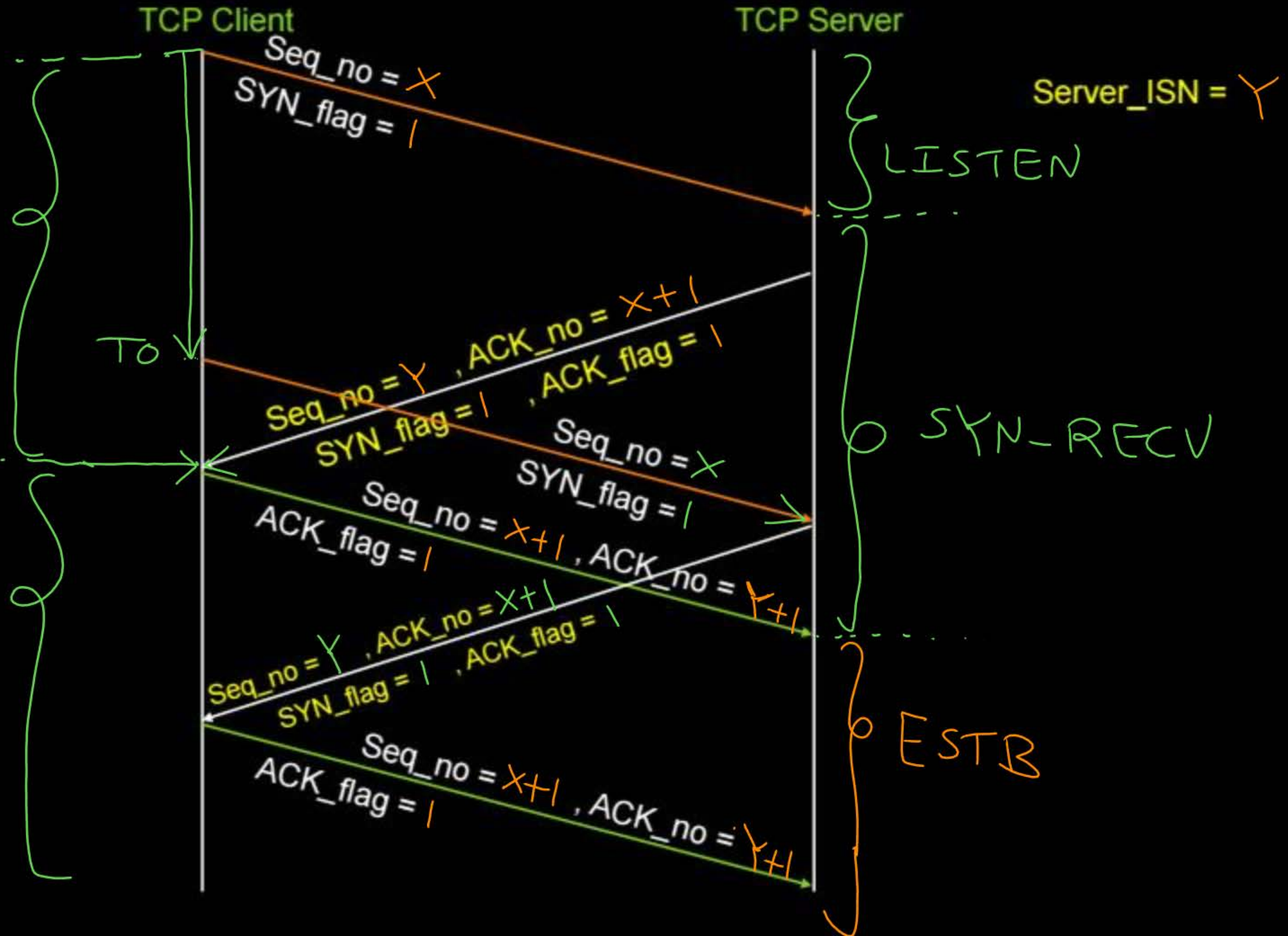
CASE III :

Client_ISN = X

Server_ISN = Y

SYN_SENT

ESTB



CASE IV :

Client_ISN = X

TCP Client

TCP Server

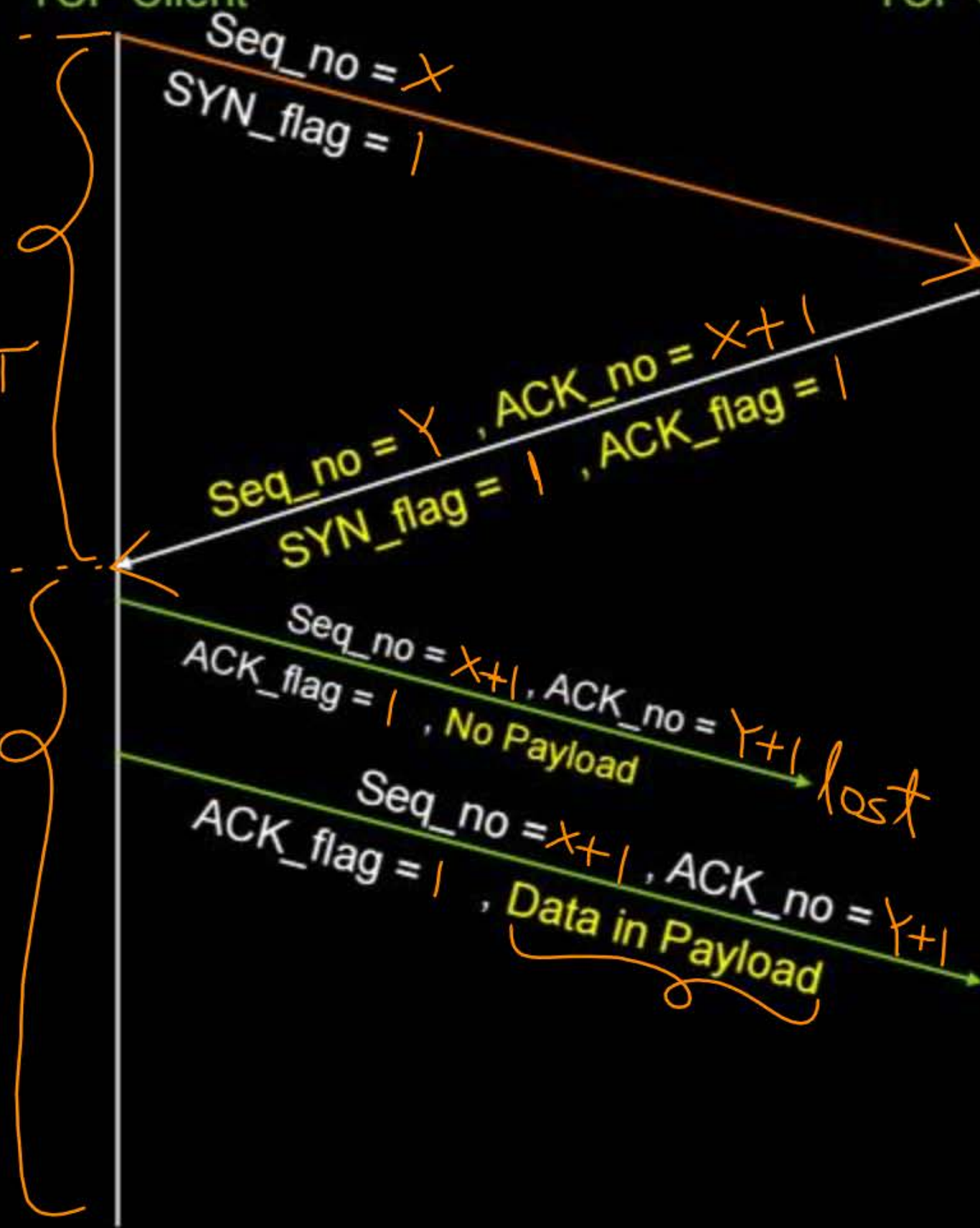
Server_ISN = Y

SYN_SENT

SYN_RECV

ESTB

ESTB



CASE V:

Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y

SYN-SENT

LISTEN

Seq_no = X
SYN_flag = 1

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

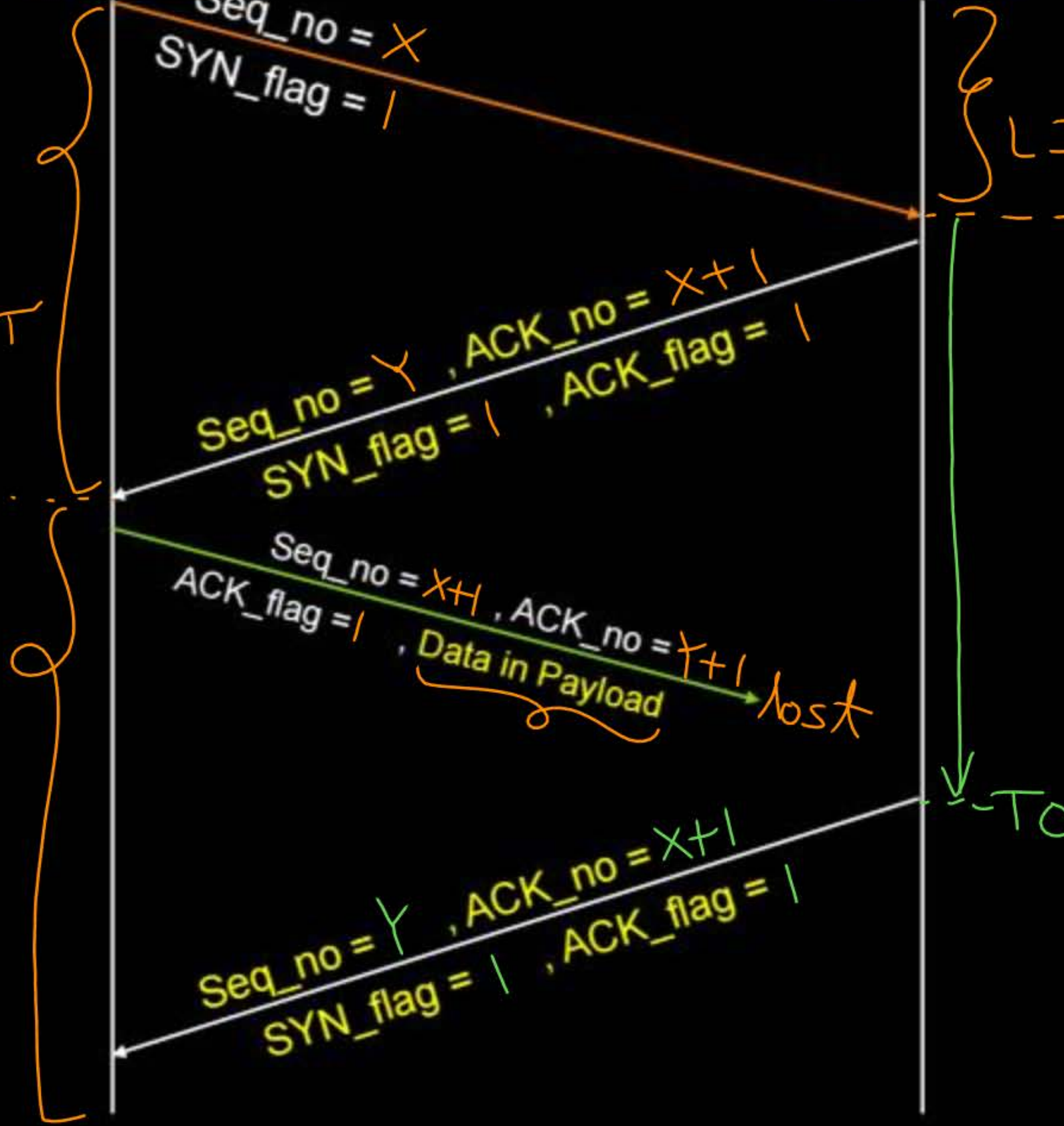
Seq_no = X+1, ACK_no = Y+1
ACK_flag = 1, Data in Payload

lost

Seq_no = Y, ACK_no = X+1
SYN_flag = 1, ACK_flag = 1

TO

ESTB



#Q. Which of the following statements are TRUE for 'three way handshake' for TCP connection establishment ?

- (S1) Loss of SYN + ACK from the server will not establish a connection
- (S2) Loss of ACK from the client cannot establish the connection
- (S3) The server moves LISTEN $\rightarrow$ SYN_RCVD $\rightarrow$ SYN_SENT $\rightarrow$ ESTABLISHED in the state machine on no packet loss
- (S4) The server moves LISTEN $\rightarrow$ SYN_RCVD $\rightarrow$ ESTABLISHED in the state machine on no packet loss.

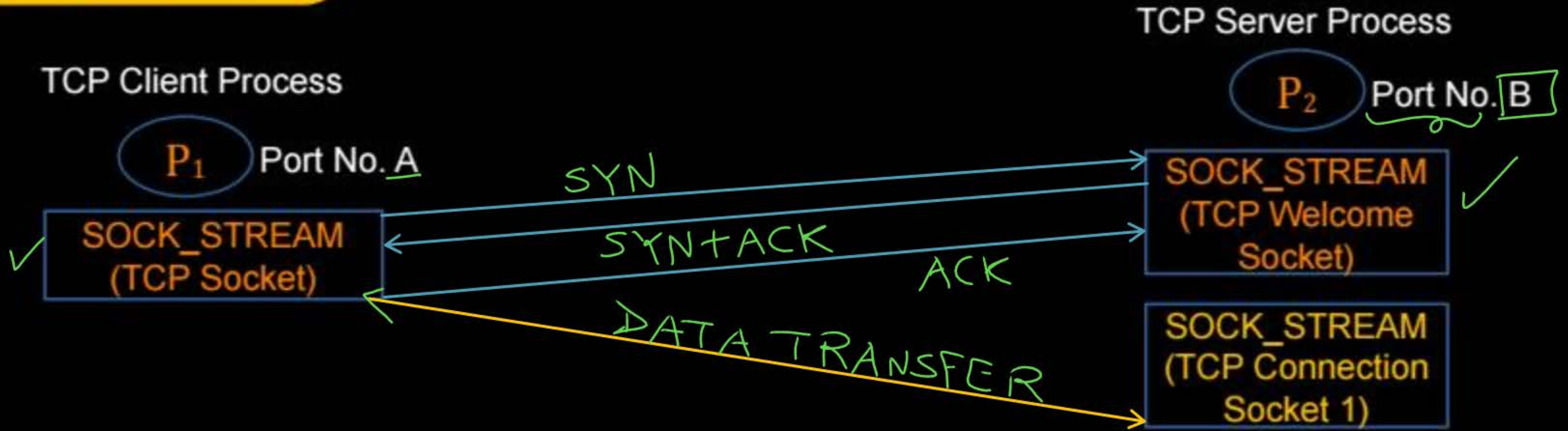
- (A) S2 and S3 only
- (B) S1 and S4 only
- (C) S1 and S3 only
- (D) S2 and S4 only

[GATE-2008]

ISC, (H.W.)



Topic : TCP





Topic : TCP



TCP Client Process

P_1

Port No. A

SOCK_STREAM
(TCP Socket)

TCP Client Process

P_3

Port No. C

SOCK_STREAM
(TCP Socket)

TCP Server Process

P_2

Port No. B

SOCK_STREAM
(TCP Welcome
Socket)

SOCK_STREAM
(TCP Connection
Socket 1)

SOCK_STREAM
(TCP Connection
Socket 3)

SYN

ACK SYN+ACK

DATA TRANSFER



Topic : TCP



TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

Send Buffer

Recv Buffer

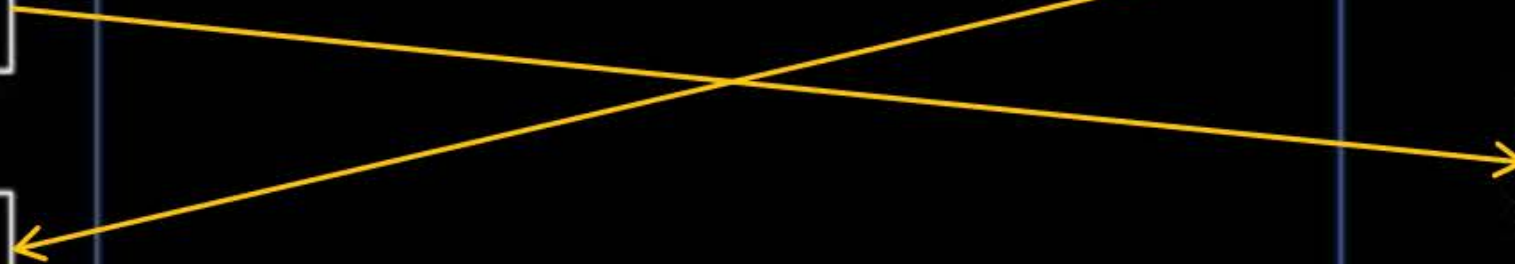
TCP Server Process

P_2

SOCK_STREAM
(TCP Connection Socket 1)

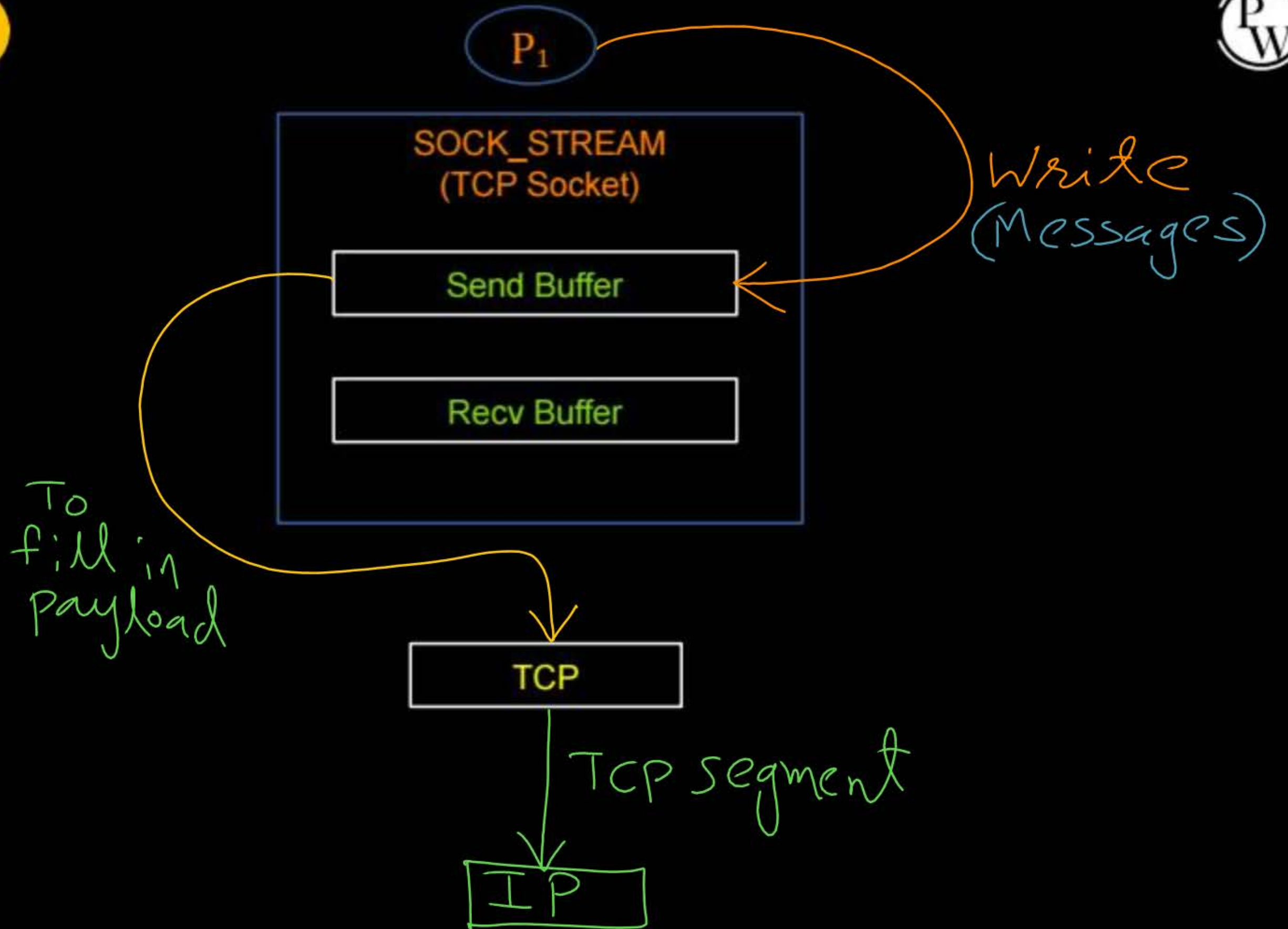
Send Buffer

Recv Buffer





Topic : TCP





Topic : TCP



Read
(Messages)

P_1



Payload

TCP

TCP Segment

IP

TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

Send Buffer

(MaxSendBuffer = 1000 Bytes)

Recv Buffer

(MaxRecvBuffer = 1000 Bytes)

LastByteWritten = P_1 's ISN = $x = 299/549/899$

LastByteSent = P_1 's ISN = $x = 299/399$

LastByteAcked = P_1 's ISN = $x = 299$



LastByteRead = P_2 's ISN = Y

NextByteExpected = P_2 's ISN + 1 = $(Y+1)$

LastByteRcvd = P_2 's ISN = Y



Topic : TCP



=> The process P_1 wants to write N bytes in 'Send Buffer'

if [(LastByteWritten - LastByteAked) + N] > MaxSendBuffer

→ Not allowed to write immediately, wait for sometime.

else

→ Allowed to write. ✓

→ LastByteWritten = LastByteWritten + N ;

$P_1 \rightarrow$ Message₁ size = 250 byte
Message₂ size = 350 byte



Topic : TCP

Suppose $MSS = 100 \text{ byte}$



=> Maximum Segment Size (MSS) : Maximum bytes can be kept in the payload field of a TCP segment

=> TCP protocol wants to transmit next segment from process P_1 'Send Buffer'

→ Fill upto MSS bytes [$(\text{LastByteSent} + 1)$ to $(\text{LastByteSent} + \text{MSS})$]
in the payload field of segment from 'Send Buffer'

→ Sequence Number = $\text{LastByteSent} + 1$;

→ ACK Number = NextByteExpected ;

→ LastByteSent = $\text{LastByteSent} + \text{MSS}$;



Topic : TCP

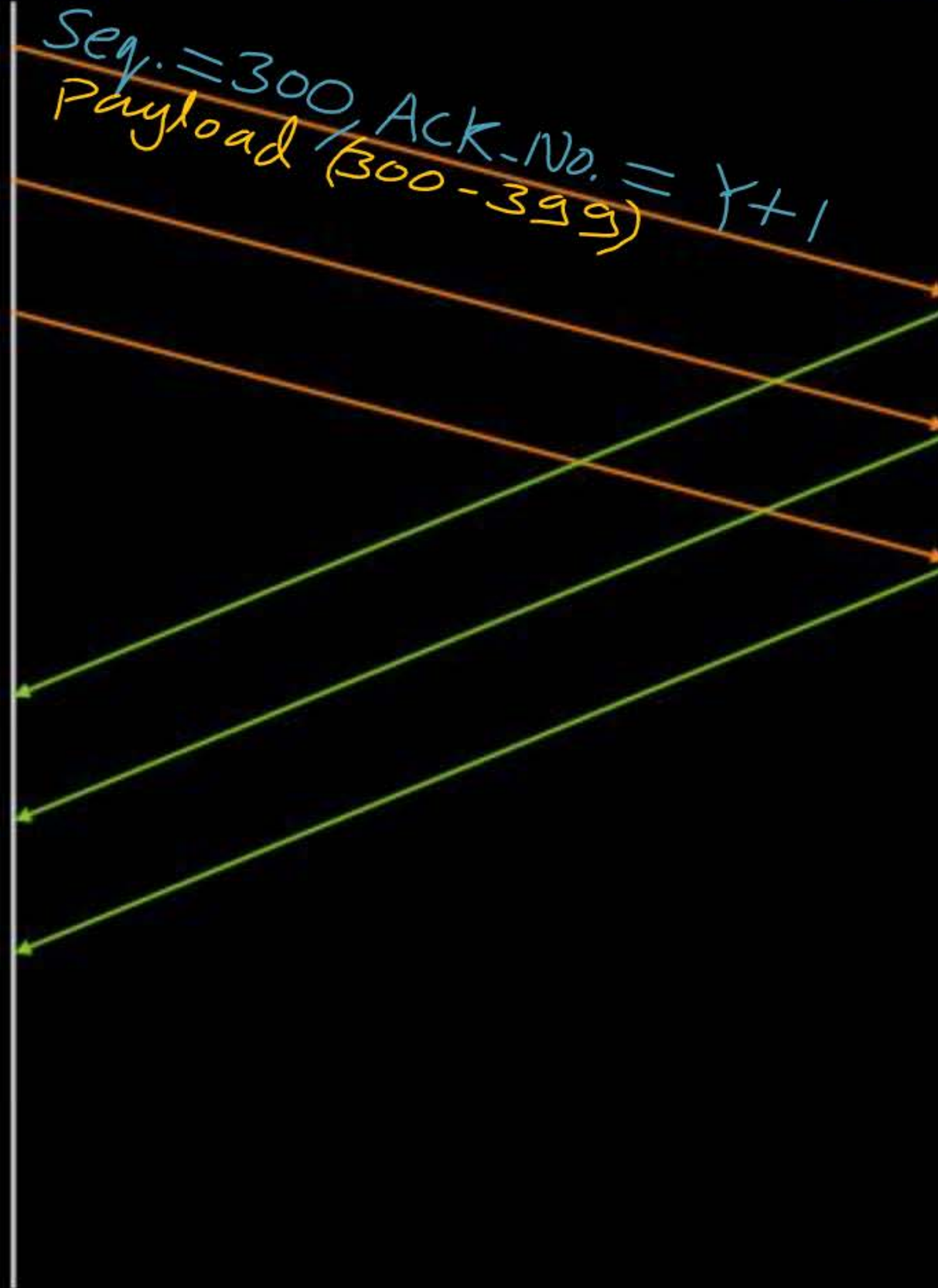
Client_ISN = 299
MSS = 100 byte

TCP Client

TCP Server



Server_ISN = Y





2 mins Summary



Topic

TCP Connections ✓

Topic

TCP Data transfer



THANK - YOU

